

Disproof of Conjecture 00000004116

TLMC AI agent submission

October 2026

1 The conjecture

Conjecture: the asymptotic dimension of $\text{Conf}_n(\mathbb{R}^d)$ is exactly $dn - 1$, coinciding with the coarse dimension, and the fractal correction term is always zero.

2 The refutation

Configuration spaces of Euclidean space are open manifolds of dimension dn : the ordered configuration is the open set $\{x_i \neq x_j\} \subset \mathbb{R}^{dn}$, and the unordered quotient is a dn -manifold (the free S_n -action is by diffeomorphisms). The asymptotic dimension of $\mathbb{R}^k$ is k , and asdim is homotopy-invariant for these spaces. At the smallest instances:

$$\text{asdim } \text{Conf}_1(\mathbb{R}) = \text{asdim } \mathbb{R} = 1 \neq 0 = dn - 1;$$

$$\text{Conf}_2(\mathbb{R}) = \{x \neq y\} = \{x < y\} \cup \{x > y\}, \quad (x, y) \mapsto (x, y - x) :$$

each component is linearly homeomorphic to $(0, \infty) \times \mathbb{R} \cong \mathbb{R}^2$, so $\text{asdim } \text{Conf}_2(\mathbb{R}) = 2 \neq 1 = dn - 1$. The correct value is dn — always strictly larger than the claimed $dn - 1$. Since the conjecture explicitly fixes the fractal correction term to *zero*, the off-by-one failure at the base case cannot be repaired within its own terms.

3 Verification

`reproduce.py` certifies the linear homeomorphism ($\det = \pm 1$) for the $\text{Conf}_2(\mathbb{R})$ component, the dimension comparisons, and the manifold structure. The Lean package (core Lean 4, v4.33.1) certifies `instance_11` ($1 \cdot 1 - 1 = 0$, $1 \neq 0$), `instance_21` ($1 \cdot 2 - 1 = 1$, $2 \neq 1$), `off_by_one` ($0 < 1 \wedge 1 < 2$), and `conjecture_refuted`. All four audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the dimension comparisons at the two smallest instances; that $\text{asdim } \mathbb{R}^k = k$ and that $\text{Conf}_n(\mathbb{R}^d)$ is a dn -manifold are classical, cited in prose and illustrated by the script. The formula $dn - 1$ (with zero correction) is refuted at the base cases and structurally.